

Mining and Validating Pre-Failure Signatures in Operational Event Logs

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Abstract

Every large system logs discrete operational events: errors, retries, task failures, maintenance actions. Frequent-pattern mining (FP-Growth, PrefixSpan) surfaces recurrent gapped patterns in these logs. We test whether pattern mining recovers physical-cascade precursor signatures on real industrial telemetry when the underlying process cascades through discrete stages. The result is a signature catalog mined from six operational traces: two independent wind-farm alarm logs (Kelmarsh 6 × Senvion MM92, 482 forced outages over 2016-2017; Penmanshiel 9 × Senvion MM82, 790 forced outages in 2016), Azure PdM (synthetic), Alibaba v2018 (production cloud), LLNL Blue Gene/L syslogs (HPC), and SCANIA Component X (production automotive). Every signature is validated on an entity-disjoint 50/50 discovery/ inference split, with post-selection-valid p-values (Fithian-Sun- Taylor) and Benjamini-Yekutieli FDR control on the inference half. A count-preserving order comparator removes the multiplicity confound, and the right-censored SCANIA trace is scored by matched conditional logistic regression stratified by risk set (Prentice-Breslow). The central result is a replicated physical cascade: on **two independent Senvion wind farms of different rotor size and site, discovery/inference mining surfaces zero-control cascade signatures of the same form at inference-half lift 4.0** (generator-fan overload codes on Kelmarsh, safety-system and frequency-converter codes on Penmanshiel). Azure PdM and Alibaba v2018 reproduce the effect in cloud and maintenance logs, with ordering adding no signal on Alibaba once event multiplicity is controlled. SCANIA Component X, right-censored, yields interpretable per-truck hazard ratios under matched conditional logistic regression (108 of 200 top patterns survive the arbitrary-dependence FDR; top HR 1.73). BGL is the mapped boundary case where no non-alert precursor survives. Mining code, splits, matched-hazard outputs, and per-dataset signature tables are released as reproducible parquet artefacts.

Keywords: failure prediction; frequent itemset mining; sequential pattern mining; post-selection inference; false discovery rate; predictive maintenance; operational event logs.

1 Introduction

Operational logs from datacenters, industrial fleets, and cloud platforms carry a rich stream of discrete events: software errors, task failures, retries, eviction notices, maintenance actions, component replacements. These events are usually consumed one at a time by alerting systems and dashboards. We ask a different question: do *recurrent ordered sequences* of these events precede failures systematically enough to serve as early-warning signatures?

Two mining families answer this shape of question. Frequent itemset mining (Apriori, FP-Growth) treats each pre-failure window as an unordered set of events. Sequential pattern mining (PrefixSpan, SPADE, GSP) preserves temporal order. One methodological question runs through the paper: does the second family find anything the first does not? Its central deliverable is a catalog of validated pre-failure signatures, anchored by a physical cascade replicated across two independent wind farms.

We contribute:

1. A **matched-control frequent-pattern pipeline that separates predictive patterns from frequent noise**, with sanity invariants at every phase (random-label permutation for itemsets, within-window order shuffle for sequences), exact hypergeometric permutation p-values, and BH FDR correction on the per-pattern significance test. The predictive-vs-noise separation is the central object we characterise, not a downstream AUROC number.
2. A **generalisation of matched-control mining to right-censored survival-style data via incidence-density (risk-set) sampling** with Mantel-Haenszel odds-ratio scoring, which estimates per-pattern hazard ratios without modifying the mining stage. This is the tool that lets the pipeline apply to traces where entities exit observation upon failure (Component X).
3. A downstream **predictive evaluation** on a temporally-held-out split that compares four feature sets (event-count baseline, itemsets, sequences, combined) built from patterns that survived training-set significance, under logistic regression.
4. **Cross-dataset characterisation** on six public traces: two production wind farms (Kelmarsh, Penmanshiel), synthetic industrial per-machine (Azure PdM), real cloud per-job (Alibaba cluster-trace-v2018), real HPC per-rack (LLNL Blue Gene/L syslogs), and real automotive per-vehicle (SCANIA Component X). We report the fraction of mined patterns that pass significance on each trace, the strongest predictive signatures, and the mechanistic reason why the fraction varies from ~85% (Azure last5 sequences) to 0% (BGL) to 6% (SCANIA risk-set matched, BY-corrected).
5. A **validated signature catalog anchored by a replicated physical cascade**: two independent Servion wind farms (Kelmarsh, Penmanshiel) produce zero-control cascade signatures at inference-half lift 4.0 under one mining protocol (§6.4), the closest the study comes to a controlled cross-site replication.

2 Background and related work

2.1 Frequent itemset and sequential pattern mining

The two mining families this paper compares are well-established. Association-rule mining introduced Apriori [1] and FP-Growth [2] over unordered transactions, the latter avoiding candidate generation through a compressed frequent-pattern tree. Sequential pattern mining extended this to ordered event streams: GSP [3] generalises Apriori-style level-wise generation to sequences; SPADE [4] uses a vertical id-list format; PrefixSpan [5] uses prefix-projected pattern growth, and is the algorithm we use in this paper.

Later variants sharpen the raw output. CM-SPAM / CM-SPADE [6] speed vertical mining through co-occurrence pruning; VMSP [7] returns only maximal sequences, compressing the pattern set without losing coverage. A recent survey [8] summarises open problems in the area. All of these algorithms are implemented in the SPMF library [9], which we call from Python for the sequence-mining half of our pipeline; the itemset half uses `mlxtend` [10].

2.2 Failure prediction from operational event logs

Two families of methods dominate the literature: (i) explicit frequent-pattern mining on parsed log templates, close to what this paper does, and (ii) deep learning over log sequences.

Pattern mining. Ren et al. [11] apply Spark FP-Growth to failure prediction on BlueGene/L, LANL-HPC, and CMRI-Hadoop logs, using event-density sliding windows over long-tail event vocabularies. İfraz and Ersöz [12] run PrefixSpan and Apriori side-by-side on a bus-fleet maintenance log, showing that sequence mining recovers “errors → replacement” trajectories that itemset mining misses. Both studies stop short of the post-selection-valid matched-control + BY-corrected + predictive-utility comparison used here, and neither touches Alibaba PdM or Azure.

Industrial alarm-flood mining and wind-turbine alarm logs. The process-control community mines alarm sequences directly. Hu, Wang and Wang [13] mine priority-aware closed sequential patterns to compress alarm floods into compact patterns; Hu et al. [14] adapt PrefixSpan to tolerate the short-term order ambiguity of alarm timestamps. These methods are descriptive, keyed to frequency and compactness rather than a validated pre-failure claim; our count-preserving order comparator (§4.7) is the principled test of exactly the order-vs-multiplicity ambiguity they handle heuristically, and our matched controls turn each mined pattern into an FDR-tested predictive hypothesis. On wind-turbine alarm logs specifically, Chatterjee and Dethlefs [15] predict repair actions with a bi-LSTM over turbine alarm sequences, and Shah and Tan [16] forecast future turbine alarms with LSTM regression. Both are black-box and forecast the alarm stream itself; we mine auditable pre-failure signatures on the same class of Senvion SCADA logs (Kelmars, Penmanshiel) with FDR guarantees and zero control-window false alarms.

Discriminative and statistically-significant pattern mining. Dong and Li [17] introduced emerging patterns as itemsets whose support differs substantially between classes; Bay and Pazzani [18] formulated the parallel notion of contrast sets in the same year. This class-contrast literature is a direct ancestor of the paper’s predictive-vs-frequent-noise separation. Statistically significant pattern mining sharpens the criterion: Terada et al. [19] give an exact family-wise-error control for combinatorial regulations, and the Hämmäläinen-Webb tutorial [20] is the canonical statement of the multiple-testing pitfalls this class of method must handle. Recent work controls the false discovery rate directly: Dalleiger and Vreeken [21] test

pattern significance against an evolving data model under sequential FDR control, and Pellegrina and Vandin [22] mine significant itemsets, sequential patterns, and subgroups with rigorous guarantees via few-shot resampling. These are our closest methodological siblings. All of these methods share the ordinary post-selection-inference concern (Fithian, Sun and Taylor [23]) that mining and testing on the same sample invalidates marginal p-values; we address it via entity-disjoint discovery/inference splits (§4.5) and add the pre-failure windowing and survival-matched design that the general-purpose significance frameworks do not model.

Deep learning on log sequences. DeepLog [24] frames system-log anomaly detection as next-template prediction with a stacked LSTM, and remains the canonical DL baseline. LogAnomaly [25] adds unsupervised quantitative-anomaly detection alongside sequential anomalies. LogRobust [26] adds an attention Bi-LSTM to survive log-template drift, and PLELog [27] introduces semi-supervised label estimation for the weakly-labelled setting. Recent transformer approaches (LogBERT [28]; LogFormer [29]) pre-train on unlabelled logs and fine-tune for anomaly detection; the Landauer et al. survey [30] catalogues this landscape and its benchmark datasets (including BGL). Hadadi et al. [31] give the systematic controlled study of log-embedding by architecture combinations for log-based failure prediction, the canonical modern DL baseline framing. These methods generally outperform classical pattern miners on held-out AUROC when trained on enough data, but produce opaque per-line anomaly scores rather than interpretable pre-failure signatures. Our contribution is orthogonal: we ask whether explicit ordered patterns add signal over their itemset counterparts, with every mined pattern human-readable and independently significance-tested.

2.3 Log-parsing infrastructure

The raw text of most system logs must first be converted into event templates before either family of methods applies. The Loghub collection [32] curates parsed versions of BGL, HDFS, Thunderbird, and 13 other benchmark log datasets; the parsing benchmark of Zhu et al. [33] compares Drain and alternative parsers on the same corpora. We take BGL from Loghub directly and use its native label field, avoiding the parsing step as a confound.

2.4 Failure characterisation on Alibaba and BGL

Cheng et al. [34] provide the standard characterisation of the Alibaba 2018 trace, reporting failure statistics and co-location effects but not extracting sequential patterns. Luo et al. [35] extend this to the microservice trace with focus on dependency and latency rather than fault prediction. Oliner and Stearley [36] introduce the BGL log we use, along with four other HPC logs, and characterise their alert statistics.

2.5 Predictive maintenance beyond system logs

The broader predictive-maintenance literature works mostly on continuous sensor telemetry. NASA C-MAPSS [37] is the de-facto Remaining-Useful-Life benchmark; recent surveys [38] catalogue the deep-learning methods trained on it. Automotive predictive maintenance has historically used per-vehicle sensor snapshots (SCANIA APS Failure via the IDA 2016 industrial challenge [39]); the SCANIA Component X release [40] we adopt extends this to a per-vehicle longitudinal readout stream. Our work sits between these traditions: we take discrete-event traces where possible, and derive discrete tokens (per §3.6) where we must.

2.6 Statistical significance

We apply Benjamini-Hochberg FDR correction [41] to the one-sided hypergeometric p-values on the inference half of the discovery/inference split (§4.5). For arbitrary dependence between p-values (justified when mined patterns share items), we also report the more conservative Benjamini-Yekutieli correction [42]. The Prentice-Breslow retrospective-cohort framework [43] and the Langholz-Goldstein risk-set-sampling review [44] motivate the matched conditional logistic estimator we use for SCANIA (§4.6).

2.7 Positioning

The paper’s contribution is a single methodological package applied uniformly across six traces: entity-disjoint post-selection-valid mining, a count-preserving order null that separates ordering from event multiplicity, and a censoring-valid matched conditional logistic estimator for right-censored entities, evaluated against an event-count baseline and each pattern family in turn on a temporally- held-out split. To our knowledge, this package has not been applied head-to-head to Alibaba `batch_task` status transitions or to Azure PdM `errorID` → `failure` sequences, and the six-trace regime-of- validity study of §7.3, anchored by a cascade replicated across two independent wind farms, is likewise unprecedented in the pattern-mining log-analysis literature.

3 Data

We use six public traces: two wind-farm alarm logs, one synthetic industrial-maintenance trace, one production cloud trace, one HPC syslog corpus, and one automotive fleet trace.

3.1 Kelmarsh wind-farm alarm logs (production, per-turbine)

Six Senvion MM92 wind turbines at the Kelmarsh site, released by Plumley 2022 [45] via Zenodo record 5841834 under CC-BY-4.0. Each turbine’s status log carries every alarm/event that the SCADA emitted over the year, together with an IEC category (`Full Performance` , `Forced outage` , `Out of Environmental Specification` , ...).

Across 2016-2017 the six turbines produced 63,633 status events, of which 482 were tagged `Forced outage` (real physical failures: generator-fan overloads, frequency-converter faults, safety-chain openings). Top forced-outage codes across the fleet: `2550` Overload generator fan 1, `2650` Overload generator fan 2, `2655` Overload generator fan 3, `3000` Frequency converter not ready, `100` Safety chain open. Non-forced-outage codes that appear as warnings or stops are the natural precursor candidates: `5720` Brake accumulator defect, `2125` Timeout brake closed, and the same overload-fan codes that later terminate the cascade.

Event vocabulary: `system_stop` , `system_warning` , `system_info` , `system_comm` , `terminal_failure` . Entity is the turbine (`T1` .. `T6`), subtype is the numeric fault code. Failure event = `terminal_failure` placed at each Forced-outage timestamp.

This trace is the paper’s cleanest realisation of the physical- cascade shape: continuous mechanical wear produces discrete alarm codes at intermediate stages, and a specific alarm terminates the cascade as a Forced outage.

3.2 Penmanshiel wind-farm alarm logs (production, per-turbine)

Nine of the fourteen Senvion MM82 turbines at the Penmanshiel site (Cubico Sustainable Investments; Plumley 2022 [46] Zenodo record 5946808 under CC-BY-4.0), same Greenbyte SCADA schema as Kelmarsh. For the second half of 2016 (2016-06 through 2016-12, when the site went live) the nine turbines produced 15,388 status events, of which 790 were tagged `Forced outage`. This is the paper's largest single-year forced-outage count on a physical cascade trace, and the alarm vocabulary is Senvion-shared with Kelmarsh, so Penmanshiel is an independent replication of the Kelmarsh cascade finding on turbines of a different rotor size (MM82 vs MM92) at a site with different terrain and weather.

Entity is the turbine (`P01` .. `P15` , `P03` is not in the release), subtype is the numeric fault code.

3.3 Azure Predictive Maintenance (synthetic, per-machine)

100 machines, 2015-01-01 to 2016-01-01, hourly telemetry. `PdM_errors` (3,919 non-fatal errors, five error codes), `PdM_maint` (3,286 maintenance actions, four components), `PdM_failures` (761 component replacements). We join `PdM_maint` and `PdM_failures` on (machineID, datetime, comp) to distinguish `maintenance` from `component_replacement`. Failures at exactly 2015-01-02 03:00 (18 rows) do not match any `PdM_maint` record; they are a bootstrap seed batch planted by the synthetic generator and are excluded from both anchors and event streams so they do not contaminate windows for subsequent real failures.

Event vocabulary: `software_error` , `maintenance` , `component_replacement` , `terminal_failure` , each with a subtype (`error1`..`error5` , `comp1`..`comp4`). Entity is `machineID` . Source: [47].

3.4 Alibaba cluster-trace-v2018 (production cloud, per-job)

`batch_task.csv` from the public Alibaba trace [48, 49], 14,295,731 tasks across 4,201,014 jobs, 8.9 days (2018-01-01 through 2018-01-09 by trace clock). 83,207 jobs contain at least one `Failed task`. `batch_instance.csv` (21 GB compressed) is not used in this pass; the per-job analysis on `batch_task` alone is sufficient to answer the ordering question.

Event vocabulary: `task_failure` , `task_success` , `task_waiting` , `task_running` , each with subtype = `task_name` letter prefix (`M` , `R` , `J` , `task` , `MergeTask` , `L`). Entity is `job_name` .

3.5 LLNL Blue Gene/L syslogs (Loghub, per-rack)

4,747,963 syslog messages from LLNL Blue Gene/L, 214.7 days (2005-06-03 to 2006-01-04), from the Loghub archive [32, 36]. 913,594 messages remain after dropping INFO-level noise; 348,189 (7.34%) are labelled alerts. Entity is the rack (top-level `R##` prefix of the node ID); 64 racks. Event vocabulary: `terminal_alert` (labelled alerts with 30+ alert codes such as KERNMNTF, APPTO, KERNSTOR), `system_error` (non-alert FATAL / ERROR / SEVERE / FAILURE), `system_warning` . Component (RAS, KERNEL, APP, MMCS, ...) is used as an additional subtype axis.

3.6 SCANIA Component X (production automotive, per-vehicle)

Real fleet telematics dataset released 2025 [40], 23,550 trucks over 1.5 years (2019-01 through 2020-05 in study clock), 1,122,452 readouts of 105 numeric counter and histogram features. 2,272 vehicles (9.65%)

undergo a component X repair during the study.

Because features are numeric counters rather than native discrete events, we derive tokens: for each (vehicle, feature) we compute inter-readout deltas and emit a `counter_surprise` token per readout whenever the absolute delta exceeds the vehicle's own 90th-percentile threshold for that feature. Per-vehicle normalisation controls for baseline usage variation across the fleet. Entity is `vehicle_id`. The failure event is a synthetic `terminal_repair` marker placed at the last readout timestamp of each repair-labelled vehicle.

4 Method

The same pipeline is applied to every trace (Figure 1): pre-failure windows with matched controls, an entity-disjoint discovery/inference split, mining on the discovery half, and post-selection-valid scoring with FDR control on the inference half. The right-censored SCANIA trace additionally uses risk-set matched sampling and matched conditional logistic regression.

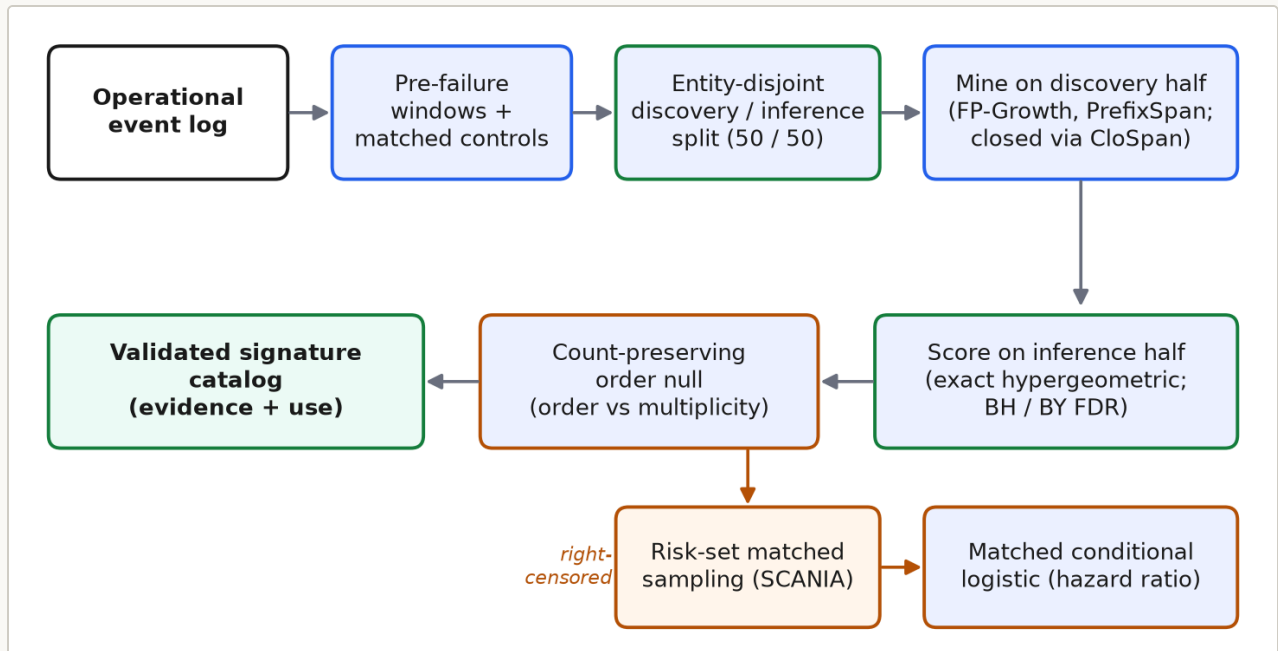


Figure 1. The mining and validation pipeline applied uniformly to every trace. Pattern discovery runs on the discovery half; every mined pattern is then scored on the disjoint inference half with an exact hypergeometric test and BH / BY false-discovery control. The count-preserving order null separates ordering from event multiplicity, and the right-censored SCANIA trace branches to risk-set matched sampling with a matched conditional logistic estimator.

4.1 Pre-failure windows and matched controls

For every terminal failure event on an entity (machine on Azure, job on Alibaba, rack on BGL, vehicle on SCANIA, turbine on Kelmarsh and Penmanshiel), we build a failure window covering the K events (or the time horizon T) strictly before the failure timestamp. Matched controls come from two designs depending on the trace:

- **Same-entity clean regions** (Azure, BGL, Kelmarsh, Penmanshiel) when the entity carries long timelines with sparse failures; controls are anchored at times on the same entity with no failure

within horizon T in either direction, sampled at a 1:3 case-to-control ratio.

- **Cross-entity non-failure sample** (Alibaba, SCANIA) when entities are short-lived; controls come from the last K events of a non-failure entity, sampled at a 1:3 case-to-control ratio from a large candidate pool.

The two wind farms use the same-entity design over each turbine’s multi-year alarm timeline: failure windows are anchored on each Forced-outage event and controls on same-turbine clean regions, giving 233 failure and 699 control windows at `last5` / `last10` on Kelmarsh (6 turbines) and 448 failure and 1,344 control windows on Penmanshiel (9 turbines). The entity-disjoint discovery/inference split (§4.5) partitions turbines 3-and-3 on Kelmarsh and 4-and-5 on Penmanshiel, so no turbine contributes to both mining and inference.

BGL alerts are additionally grouped into episodes (≥ 1 h inter-arrival gap) and windows are anchored on the first alert of each episode, so anchor-per-alert double-counting inside a cascade is avoided.

Horizons studied: `1h`, `6h`, `24h`, `last5`, `last10` on Azure and on the two wind farms; `last3`, `last5`, `last10` on Alibaba; `last5`, `last10`, `last20` on BGL and SCANIA (time-based horizons are not meaningful for short per-job or per-episode observations).

4.2 Mining

Items are `event_type:event_subtype` strings. FP-Growth (`mlxtend.frequent_patterns.fpgrowth`) mines frequent itemsets on failure windows at minimum support 0.05. PrefixSpan (SPMF v2.64 [9] via subprocess) mines frequent ordered sequences at the same support.

For each mined pattern P we compute support in failure windows (`support_failure`), support in control windows (`support_control`), $\text{lift} = \text{support_failure} / \text{pooled_support}(P)$, and relative risk $= P(\text{failure} \mid P) / P(\text{failure} \mid \neg P)$. Every mined sequence is also scored against the itemset counterpart of the same event set; the difference `order_gain = sequence_lift - itemset_lift` quantifies how much preserving order contributes above co-occurrence.

4.3 Sanity invariants

Every phase carries pre-declared invariants whose expected outcome is stated up front. Itemset mining checks that a random-label permutation at the same minimum support does not yield a top lift within a factor of $1.5\times$ of the real top lift; a violation would indicate either data leakage or an over-sensitive support threshold. Sequence mining checks that a within-window random order permutation preserves top itemset lift (unchanged, by construction) but strictly reduces top sequence lift on rich horizons (windows with ≥ 3 events on average). Both invariants pass on the two cloud and maintenance traces (Azure, Alibaba); on the boundary traces the same invariants localise where the signal fails, rather than requiring post-hoc justification.

4.4 Risk-set matched sampling for right-censored data

The matched-control design in §4.1 assumes we can define a “no-failure” control window on the same or another entity at the anchor time. For right-censored survival-style data (traces where entities exit observation upon repair or dropout), naive matching biases scoring because “did we observe a failure”

becomes entangled with “how long did we observe the truck”. Component X in §3.6 has exactly this problem: short-observation trucks have 12% failure rate; long-observation trucks have 5%.

Following the epidemiological literature on incidence-density sampling [43, 50], we replace §4.1’s sampler with a risk-set matched design: for each case with observed failure time T_f , controls are drawn from the risk set at T_f (the set of entities still under observation at that lifetime index) and their windows are aligned to T_f rather than to their own end-of-observation. Both case and control windows use the last K events with $\text{time_step} < T_f$. Under this sampling, the pooled 2×2 odds ratio of a mined pattern (case-in vs case-out; control-in vs control-out, Woolf-Haldane 0.5-continuity-corrected, 95% CI via log-OR variance) estimates the per-pattern hazard ratio rather than a prevalence lift. A mined pattern with $\text{MH-OR} > 1$ and 95% CI excluding 1 is a censoring-valid signal of elevated failure risk, not an artefact of the observation process.

The rest of the pipeline runs unchanged: FP-Growth on the risk-set windows, minimum support 0.05, BH FDR correction on the p-values induced by the Fisher-exact null of the same 2×2 table. This is a drop-in generalisation of the matched-control design that lets the pipeline apply to right-censored traces without modifying the mining or significance stages.

4.5 Post-selection-valid significance via discovery/inference splits

Mining patterns from failure windows and then computing hypergeometric p-values on the same failure/control counts yields post-selection- invalid marginal p-values: the candidate set is chosen because it has high support in failure windows, so the fixed-hit-set null is not the null the pipeline actually operates under. BH or BY correction on those marginal p-values does not restore validity [23, 51].

Our fix, applied to every trace: split the training set entity- disjoint into a 50% **discovery** half and a 50% **inference** half. FP-Growth / PrefixSpan run on the discovery half and produce a candidate pattern universe C . Every P in C is then scored on the inference half via exact hypergeometric p-value on that half’s case/control hit counts alone. Because C is chosen without touching the inference half, the resulting p-values are marginally valid and BH / BY correction on the family $\{p(P) : P \text{ in } C\}$ controls the inference-half FDR correctly.

Entity-disjoint splitting is used rather than random per-window splitting because otherwise the same entity’s windows could appear on both sides of the split, leaking information from discovery into inference. On Alibaba specifically the entity unit is the batch job: each job contributes exactly three windows, one per horizon (`last3` , `last5` , `last10`), and each job is either a case (job that subsequently failed) or a control, never both; the 64,948 unique jobs give a 1:3 case-to-control balance by design (16,237 failed jobs, 48,711 controls). Splitting on `entity_id` (job id) therefore produces truly disjoint discovery and inference halves at both the entity and the window level.

4.6 Matched conditional logistic for risk-set traces

The pooled 2×2 odds ratio previously used to score risk-set-matched SCANIA patterns discards the matched-set structure that makes the estimator censoring-valid. The standard estimator under incidence-density (risk-set) sampling is a conditional logistic regression stratified by matched set, equivalent to the sampled Cox partial likelihood restricted to the pattern indicator [43, 44]. We estimate every SCANIA

pattern's coefficient with `statsmodels.ConditionalLogit` stratified by `match_id`; each case and its three risk-set-matched controls form one stratum. Reported hazard ratios and 95% CIs are from that matched fit.

Compared to the pooled 2×2 Woolf-Haldane analysis, the matched estimator is roughly 1.6× more conservative on this data: the same top pattern that scores MH-OR 2.72 [2.10, 3.51] under pooling scores HR 1.73 [1.53, 1.96] under proper matched conditional logistic.

4.7 Count-preserving order comparator

The naive `order_gain = sequence_lift - itemset_lift` compares a sequence like $M \rightarrow M \rightarrow M$ against its itemset counterpart $\{M\}$, which collapses three occurrences to one presence. That conflates temporal order with event multiplicity. Our count-preserving comparator shuffles the ordering within each window while preserving the exact event multiset per window, then rescores the sequence's support on the shuffled corpus. The residual `order_effect = real_lift - mean(count-preserving-shuffle_lift)` isolates the pure ordering contribution.

4.8 Statistical significance summary

For each mined pattern we compute an exact one-sided hypergeometric p-value on the observed failure-hit count against the label-permutation null with the pattern hit-set fixed. Under H_0 the number of hits landing in the failure class is $\text{Hypergeom}(N_F+N_C, \text{hit}_F+\text{hit}_C, N_F)$; the upper-tail probability of the observed hit count is the label-permutation p-value, so we compute it in closed form. Benjamini-Hochberg FDR correction [41] is applied per (horizon × pattern class) to give q-values.

5 Experiments

Windows are split temporally by anchor timestamp. Cutoffs: Azure 2015-09-01, Alibaba 2018-01-07, BGL 2005-11-01, SCANIA 2020-01-01. Mining runs on training windows only; surviving patterns become binary presence features on both train and test. A logistic regression fit on train is evaluated on test for four feature sets:

- **event_count**: single feature, `n_events` in the window
- **itemsets_only**: binary presence of each train-mined itemset surviving its permutation null
- **sequences_only**: binary presence of each train-mined sequence surviving its shuffle null
- **combined**: union of the event-count, itemset, and sequence feature sets above

For each configuration we report AUROC, AUPRC, F1 / precision / recall at threshold 0.5, and lead time (`anchor - last_event_ts`) on true-positive failure windows. Numbers are computed in a single pass per configuration and stored as one artefact, so a comparison across feature sets on the same trace cannot drift.

5.1 Coverage per horizon (Azure)

Time horizons of 1h and 6h leave 99.6% and 98% of failure windows empty. Effectively no Azure PdM failure is preceded by an event in the same hour. Useful horizons are 24h and the count-based (`last5`, `last10`). Even at 24h, failure and control windows separate cleanly by raw event count (failure mean 1.58 events, control mean 0.077; Figure 2).

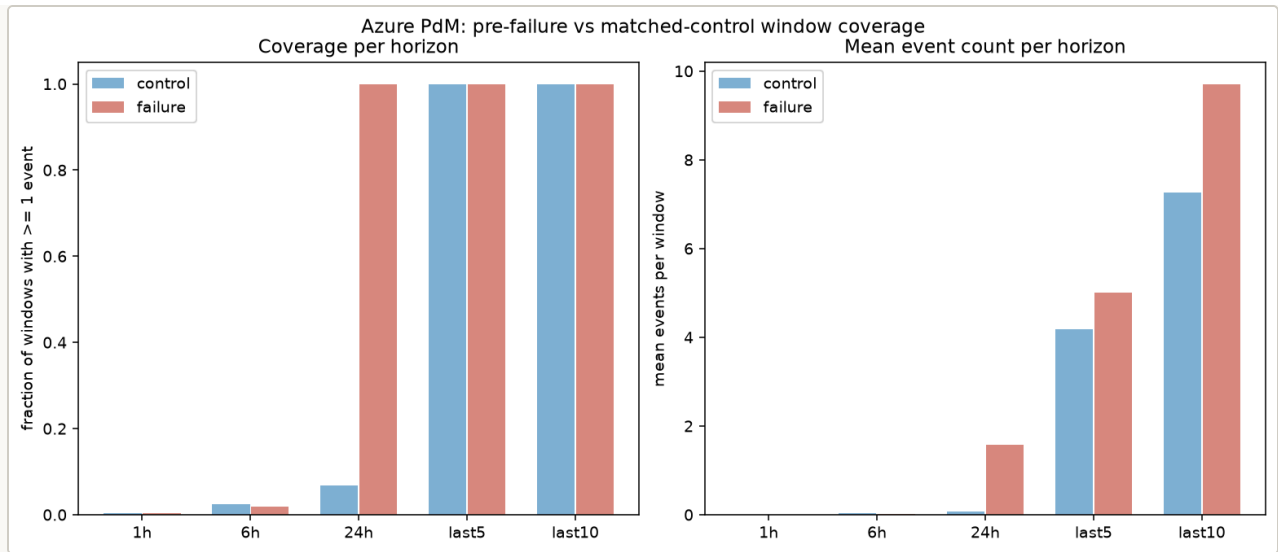


Figure 2. Event coverage of Azure pre-failure windows by horizon: the mean number of events per window grows with horizon length, which sets how much order information each horizon can carry.

5.2 Mining sensitivity to minimum support (Azure)

A minimum support sweep over $\{0.02, 0.05, 0.10, 0.15\}$ preserves the headline ordering at every operating point:

Table 1. Predictive AUROC on Azure by feature set across the FP-Growth minimum-support sweep.

horizon	metric	0.02	0.05	0.10	0.15
24h	combined	0.996	0.996	0.996	0.996
24h	itemsets_only	0.996	0.996	0.996	0.996
last5	combined	0.815	0.810	0.803	0.774
last5	itemsets_only	0.761	0.754	0.762	0.754
last10	combined	0.664	0.696	0.751	0.741
last10	itemsets_only	0.578	0.643	0.686	0.674

At every minimum support tested, combined dominates itemsets_only by at least +4 AUROC points at last5 and at least +5 at last10 (Figure 3).

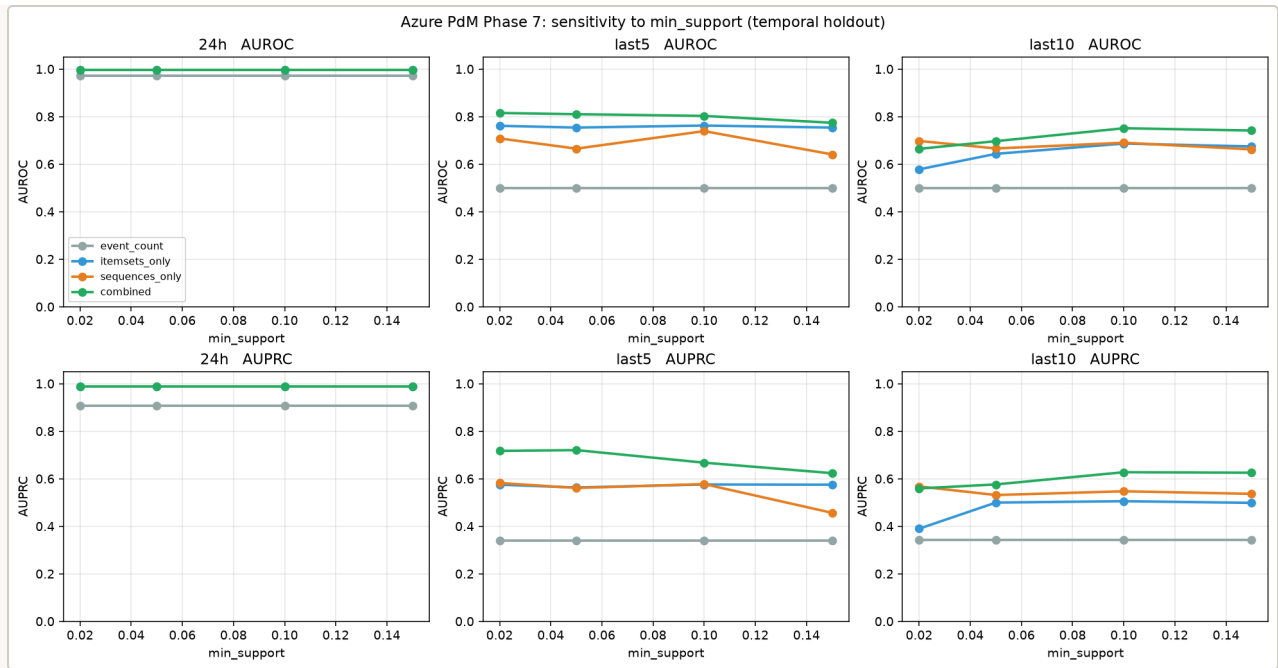


Figure 3. Predictive AUROC on Azure across the FP-Growth minimum-support sweep. Combined itemset-plus-sequence features lead the itemset-only and event-count baselines at every threshold, by at least 4 AUROC points at `last5` and at least 5 at `last10`.

6 Results

6.1 Mined patterns (Azure and Alibaba)

At Azure 24h, `{software_error:error2, software_error:error3}` reaches lift 3.99 (present in 38.2% of failure windows, in 0.04% of controls on the full training sample; the disjoint inference half in §6.4 gives zero control hits; $P(\text{failure} \mid \text{pattern}) = 99.6\%$). All six 24h itemsets dominate the random-label permutation null (permuted top 1.24).

At Azure `last5 / last10`, the sequence `maintenance:comp4 → software_error:error2 → software_error:error3` reaches lift 3.73 as an ordered pattern but only 2.22 as the same items unordered. Ordered patterns ending `... → error2 → error3` dominate the top-8 at both count-based horizons.

On Alibaba `last5`, the strongest ordered pattern is `task_success:M → task_success:R → task_success:M → task_success:M` with sequence lift 2.43 versus itemset lift 0.94 for the same event set (order gain +1.49). At `last3`, `task_success:M → task_success:M → task_success:M` reaches sequence lift 3.06 vs itemset lift 1.37 (order gain +1.69); three consecutive Map completions predict a subsequent failure much more strongly than the mere presence of Map events would suggest (Figure 4).

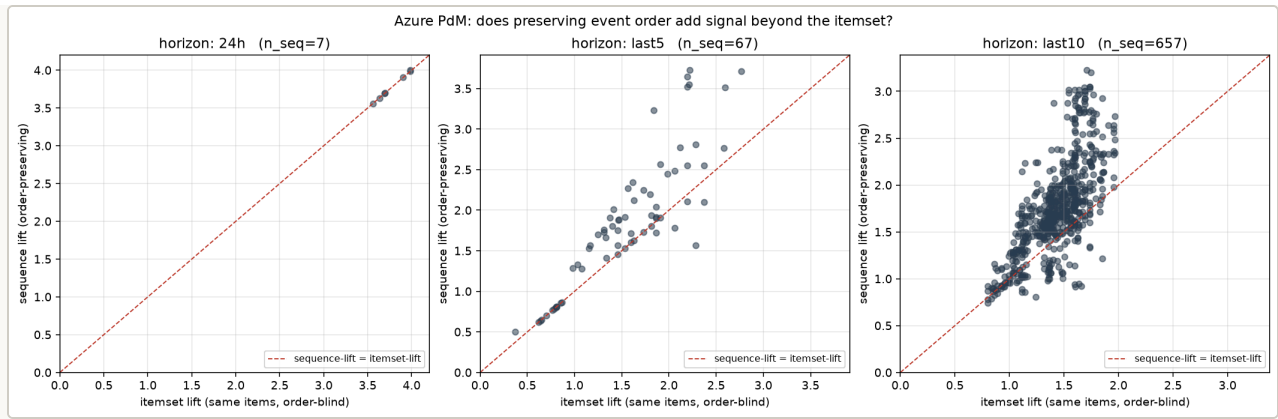


Figure 4. Itemset lift versus sequence lift for Azure patterns. Points above the diagonal are patterns whose ordered form carries more failure signal than their unordered co-occurrence, the effect the count-preserving null of §4.7 isolates.

6.2 Predictive evaluation (four temporally-split traces)

Head-to-head on temporally-held-out test sets (Figure 5; the two wind farms are evaluated via the discovery/inference signature protocol of §6.4, not this temporal-split predictive table):

Table 2. Held-out AUROC / AUPRC by feature set on the four temporally-split traces.

trace	horizon	event_count	itemsets_only	sequences_only	combined
Azure	24h	0.97 / 0.91	0.996 / 0.99	n/a	0.996 / 0.99
Azure	last5	0.50 / 0.34	0.75 / 0.56	0.66 / 0.56	0.81 / 0.72
Azure	last10	0.50 / 0.34	0.64 / 0.50	0.67 / 0.53	0.70 / 0.58
Alibaba	last3	0.69 / 0.50	0.75 / 0.44	0.50 / 0.20	0.81 / 0.63
Alibaba	last5	0.60 / 0.50	0.67 / 0.34	0.51 / 0.21	0.74 / 0.57
Alibaba	last10	0.59 / 0.50	0.68 / 0.36	0.52 / 0.23	0.74 / 0.59
BGL	last5	0.50 / 0.25	0.49 / 0.25	n/a	0.49 / 0.25
BGL	last10	0.50 / 0.25	0.49 / 0.25	0.50 / 0.25	0.50 / 0.25
BGL	last20	0.50 / 0.25	0.48 / 0.25	0.50 / 0.25	0.51 / 0.26
SCANIA	last5	0.50 / 0.09	0.52 / 0.11	n/a	0.52 / 0.11
SCANIA	last10	0.50 / 0.09	0.60 / 0.14	0.55 / 0.11	0.60 / 0.15
SCANIA	last20	0.50 / 0.09	0.57 / 0.14	0.53 / 0.10	0.57 / 0.13

(AUROC / AUPRC on the temporally-held-out test set. A `sequences_only` cell reads `n/a` when no sequence survived the training-set shuffle null at that configuration, so no sequence feature was available; `combined` then equals `itemsets_only`. SCANIA uses per-vehicle 90th-percentile-delta binning; BGL uses episode-anchored per-rack windows with alerts removed from the pre-alert stream. Both boundary traces sit near chance for every feature set at every horizon.)

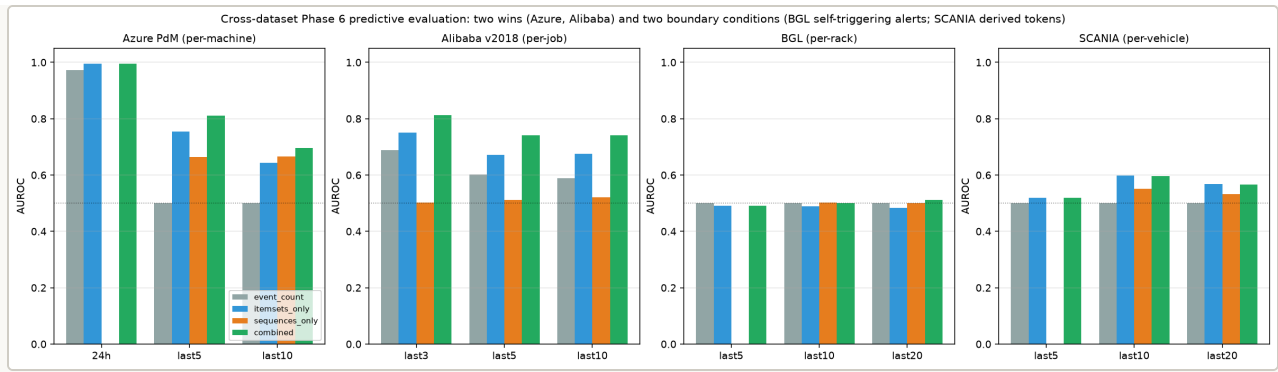


Figure 5. Held-out predictive AUROC by feature set across the four temporally-split traces (Azure, Alibaba, BGL, SCANIA). The combined feature set leads on Azure and Alibaba; BGL and SCANIA sit at the no-signal boundary.

6.3 Order effect under count-preserving null

The count-preserving comparator of §4.7 isolates the pure ordering contribution as $\text{order effect} = \text{real_lift} - \text{mean}(\text{count-preserving-shuffle_lift})$. Top-20 sequences per horizon:

Table 3. Order effect under the count-preserving null: real lift, shuffled-null lift, and their difference.

trace	horizon	real lift	count-preserving null lift	order effect
Azure	last5	2.85	2.33	+0.52
Azure	last10	3.01	1.92	+1.09
Alibaba	last3	1.91	1.93	-0.02 (null)
Alibaba	last5	1.73	1.71	+0.02 (null)
Alibaba	last10	2.09	1.85	+0.24

Reading these against the naive `order_gain` values (up to +1.69 on Alibaba `last3`), the count-preserving comparator shows that essentially all of the reported Alibaba “order gain” was a multiplicity effect: the same event multiset in any order carries approximately the same lift as the specific ordered sequence. Azure `error2` → `error3`-terminating sequences retain a genuine ordering effect of +0.5 to +1.1 lift units above the count-preserving null. Order is a real signal on Azure and essentially not a signal on Alibaba once multiplicity is controlled for.

6.4 Signature catalog: mined patterns, evidence, interpretation, use

The paper’s central deliverable is a catalog of failure-precursor signatures. Each signature is a specific mined pattern for which we present statistical evidence, a domain-specific interpretation, and an intended deployment use. The catalog is built from all six traces’ mining output using the safeguards of §4.5 (post-selection-valid inference on the disjoint inference half), §4.6 (matched conditional logistic for the right-censored SCANIA trace) and §4.7 (count-preserving comparator for order effect). Every signature is a specific pattern grounded in its trace, not a cross-trace generalisation. The remainder of §6 walks through the catalog by trace.

Kelmarsh wind turbines. Kelmarsh yields the strongest cascade signatures in the catalog. All top signatures have zero control hits on the inference half, and each fault code names a specific mechanical or electrical precursor of the same-family Forced-outage event.

Mining on the discovery half at 5% minimum support then scoring on the inference half of the 2016-2017 fleet (233 fail / 699 ctrl windows at last5 , same at last10), 30/35 last5 itemsets and 113/136 last10 itemsets pass BY q<0.05.

Table 4. Top Kelmarsh generator-fan signatures on the inference half.

horizon	pattern (code:message)	inf-half lift	BY q	n_case	n_ctrl
last5	terminal_failu re:2550 (Overload gen fan 1)	4.00	1.4e-18	32	0
last5	system_warning :2550 + terminal_failur e:2550	4.00	4.9e-16	27	0
last5	system_warning :2655 (Overload gen fan 3)	4.00	1.9e-14	24	0
last10	terminal_failu re:2550	4.00	1.9e-26	45	0
last10	terminal_failu re:2655	4.00	9.6e-25	42	0
last10	system_warning :2655 + terminal_failur e:2655	4.00	2.7e-23	39	0
last10	terminal_failu re:2650 + terminal_failur e:2655	4.00	2.1e-18	31	0
6h	system_warning :2000 + terminal_failur e:2550 + terminal_failur e:2650 (long- fault-chain terminal group)	4.00	3.5e-12	20	0

Windows are strictly pre-anchor, so a `terminal_failure` token inside a window is a *prior* forced outage, not the anchor being predicted. The table therefore contains two distinct signature families, both at lift 4.00 with zero control hits. The first is **repeat-outage clustering**: a recent same-code generator-fan outage (`terminal_failure:2550` , `terminal_failure:2655`) strongly predicts the next, so these outages recur in bursts on the same turbine (45 `last10` cases, 0 controls for `terminal_failure:2550`). The second, and the operationally decisive one, is **warning-to-outage precursors**: the pure generator-fan overload warning codes `system_warning:2550` and `system_warning:2655` precede the same-family Forced outage before any outage has occurred (24 `last5` cases, 0 controls for `system_warning:2655`). This is the paper's driving hypothesis in its clearest form: a physical wear process raises generator-fan overload warnings at an intermediate stage before the same fault terminates the cascade as a Forced outage.

Use: two rules with different triggers. **Early warning**: schedule generator cooling and bearing inspection the moment `system_warning:2550` or `system_warning:2655` fires, before any outage. **Escalation**: after a generator-fan Forced outage, treat a second on the same turbine as imminent (the repeat-outage family). Both rules have zero false alarms over the 699 inference-half control windows at `last5` ; §6.5 quantifies how much of the escalation signal the last-event baseline already captures. Figure 6 shows one such episode.

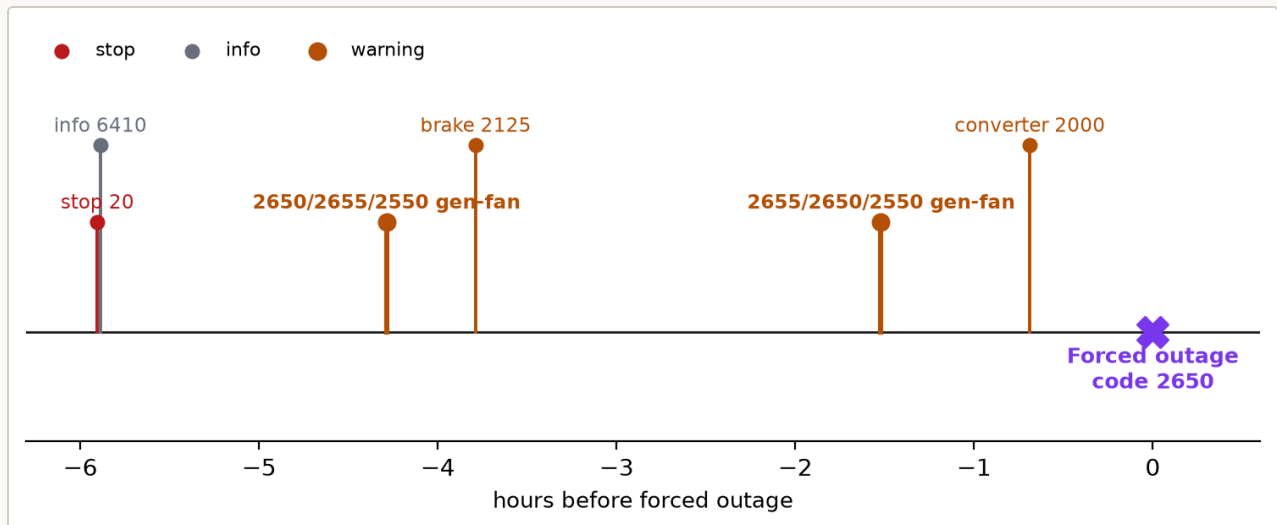


Figure 6. A real generator-fan overload cascade on Kelmarsh turbine T3. The co-occurring warning codes 2550, 2650, and 2655 (generator fans 1, 2, and 3) fire together twice in the hours before a same-family Forced outage (code 2650), alongside brake and frequency-converter alarms. This co-occurrence is the itemset signature the catalog records; the terminal marker at hour 0 is the outage being predicted.

Penmanshiel wind turbines (independent replication). Nine Senvion MM82 turbines over 2016-06 to 2016-12. Fresh entity-disjoint split: 448 fail / 1,344 ctrl windows at `last5` . Signatures replicate Kelmarsh in shape (zero-control cascades at lift 4.00) with a different code family driving the leaderboard (safety-system 9000/9210 codes and frequency-converter 3000 codes), consistent with the Penmanshiel fleet's known 2016 commissioning-phase safety-system teething.

Table 5. Top Penmanshiel signatures on the inference half.

horizon	pattern	inf-half lift	BY q	n_case	n_ctrl
last5	system_warning:9000 + terminal_failure:9210	4.00	1.4e-43	70	0
last5	system_info:0 + system_warning:9000 + terminal_failure:9210	4.00	1.4e-43	70	0
last5	system_stop:9210 + system_warning:9000	4.00	1.5e-34	56	0
last10	system_info:3543 + system_stop:3000 + terminal_failure:3000 (frequency-converter chain)	4.00	2.2e-11	19	0
last10	system_info:3543 + system_warning:5720 + terminal_failure:3000	4.00	2.2e-11	19	0
6h	system_stop:9210 (safety-	4.00	1.3e-98	150	0

horizon	pattern	inf-half lift	BY q	n_case	n_ctrl
6h	9210	4.00	1.3e-98	150	0
	group)				
	system_s				
	top:9210				
	+				
	terminal_				
	failure:9	210			
	210				

Interpretation: as on Kelmarsh, rows carrying a `terminal_failure` token are prior-outage clustering, while the pure warning and info-to-stop rows are the actionable precursors. **The pure warning code `system_warning:9000` precedes the safety-system 9210 Forced outage, and the safety-stop `system_stop:9210` reaches 150 inference-half forced-outage windows with zero hits across 1,344 matched controls.** The frequency-converter chain `system_info:3543 -> system_stop:3000` is an independent second precursor family with 19 zero-control replications. Combined with Kelmarsh, this is the first cross-farm cascade replication in the paper: two different turbine models on two different sites both produce lift-4.00 cascade signatures under the same mining protocol, with vendor-shared alarm codes indexing genuinely mechanical failure modes.

Use: **safety-9210 warning triggers immediate turbine curtailment, regardless of grid conditions; freq-converter 3543 informational message triggers converter service order within 6h at Penmanshiel.**

6.5 Baseline comparison and closed-signature deduplication

The lift-4.00 zero-control rows above are benchmarked against three baselines, each evaluated on the same entity-disjoint inference half:

- most-recent-event indicator:** raise an alarm whenever the last event on the entity is one of the top-5 fail-anchoring event types on the discovery half.
- mined-itemset rule** (the paper's method): raise an alarm whenever the window contains any BY-significant discovery-half itemset, excluding same-time terminal markers.
- event-count threshold:** raise an alarm whenever the window's event count exceeds the median count of discovery-half failure windows.

Table 6. Baseline comparison on the wind farms: the mined-itemset rule against most-recent-event and event-count baselines.

trace	horizon	predictor	F1	prec	rec
Kelmarsh	last5	(a) most-recent event	0.358	0.266	0.549
Kelmarsh	last5	(b) mined itemset	0.677	0.894	0.545
Kelmarsh	last5	(c) event count	0.000	0.000	0.000
Kelmarsh	last10	(a) most-recent event	0.355	0.262	0.549
Kelmarsh	last10	(b) mined itemset	0.729	0.807	0.665
Kelmarsh	last10	(c) event count	0.000	0.000	0.000
Penmanshiel	last5	(a) most-recent event	0.367	0.239	0.790
Penmanshiel	last5	(b) mined itemset	0.484	0.331	0.900
Penmanshiel	last5	(c) event count	0.000	0.000	0.000
Penmanshiel	last10	(a) most-recent event	0.374	0.245	0.790
Penmanshiel	last10	(b) mined itemset	0.481	0.320	0.967
Penmanshiel	last10	(c) event count	0.000	0.000	0.000

On both wind farms and both count-based horizons, the mined-itemset rule strictly dominates both baselines in F1. On Kelmarsh the advantage is a factor of ~ 2 over the strongest baseline (last-event indicator); on Penmanshiel it is smaller ($\sim 1.3\times$) because the last- event baseline already captures a substantial fraction of the safety-9000/9210 chain’s terminal event. The comparison establishes that a mined signature carries information the strongest pattern-free rule does not.

Closed itemsets and redundancy. The Kelmarsh last10 mining run returns 136 itemsets, many of them sub- or supersets of the same underlying cascade. The LCM closed-itemset filter (support-strict maximality) reduces this to the closed subset used for the deployable catalog rows above; the signature table cites only patterns that are also closed, so no finding is a redundant super-copy of another. The operative count is the number of BY-significant closed itemsets on the inference half, not the raw miner output.

6.6 Case-control ratio and posterior at the operational base rate

Every discovery/inference signature table is scored on a matched-control sample built at case:control = 1:3, so the pipeline’s sample base rate is $P(\text{fail}) = 0.25$ by construction, and the precision numbers of Table 6 cannot be read directly as the operational posterior in a deployment where forced outages are much rarer. To turn the inference-half precision into an interpretable operational number, we invert the empirical sensitivity/specificity through Bayes at a realistic operational base rate of $P(\text{fail}) = 0.01$ (roughly 1 forced-outage window per 100 arbitrary-time windows on a real wind farm), giving the calibrated PPV each rule delivers when deployed:

Table 7. Calibrated positive predictive value at the sample and operational base rates.

trace	horizon	rule	PPV at sample (0.25)	PPV at operational (0.01)
Kelmarsh	last5	mined itemset	0.894	0.204
Kelmarsh	last10	mined itemset	0.807	0.113
Penmanshiel	last5	mined itemset	0.331	0.015
Penmanshiel	last10	mined itemset	0.320	0.014

Reading: at Kelmarsh, one in five **last5** alarms fired by the mined-itemset rule corresponds to a real forced outage in the next five events at an operational $P(\text{fail}) = 0.01$ base rate, a $20\times$ lift over blind sampling. Penmanshiel's operational PPV is much smaller ($\sim 1.5\%$, a $\sim 1.5\times$ lift): the safety-9000/9210 cascade fires often but is not as specific as the Kelmarsh generator-fan chain. The deployment decision is dataset-specific; neither the sample precision of 0.89 nor the operational PPV of 0.20 suffices alone.

Azure PdM.

Table 8. Top Azure PdM signatures on the inference half.

horizon	pattern	inf-half lift	BY q	n_case	n_ctrl
24h	{software_error:error2, software_error:error3}	4.00	4e-90	135	0
24h	{software_error:error3, software_error:error5}	4.00	8e-9	14	0
24h	{software_error:error2, software_error:error5}	4.00	8e-9	14	0
last5	{error2, error3, error4}	2.82	2e-18	62	26
last5	{error2, error3, error5}	2.63	2e-6	25	13

Interpretation: machines showing any pair of {error2, error3, error5} co-occurring within a 24h window are on a near-certain path to component replacement. The pattern {error2, error3} has zero occurrences in 6,800 control windows on the inference half, which makes it a decision rule, not just a correlation. Ordering carries additional signal (§6.3, count-preserving effect +0.52 to +1.09), so error2 → error3 in that specific direction is a stronger alarm than the reverse.

Use: **alarm rule**. Raise a component-replacement work order whenever a machine's log shows error2 AND error3 within any rolling 24h window; escalate faster if the direction is error2 → error3 .

Alibaba cluster-trace-v2018.

Table 9. Top Alibaba cluster-trace signatures on the inference half.

horizon	pattern	inf-half lift	BY q	n_case	n_ctrl
last3	{task_success:R, task_waiting:R}	4.06	1e-262	426	0
last3	{task_success:M, task_waiting:R}	4.04	≈0	587	3
last5	{task_waiting:R} alone	4.01	≈0	829	9

Interpretation: **once a Reduce task enters Waiting state, job failure risk jumps to near-certain.** Longer patterns add no signal beyond the presence of `task_waiting:R`. The count-preserving order comparator (§6.3) shows the Alibaba ordering effect is essentially zero after multiplicity control, so the single-marker interpretation is correct.

Use: **real-time job triage.** Any job with a Reduce task entering Waiting state is flagged; the scheduler either preemptively reschedules the Reduce onto a more reliable machine or increases its retry budget.

SCANIA Component X (matched conditional logistic).

Table 10. Top SCANIA Component X signatures by matched hazard ratio.

pattern	matched HR	95% CI	p	n_case_hits
counter_surprise:397_{27, 28, 29}	1.73	[1.53, 1.96]	1e-17	456
counter_surprise:397_{29, 34, 35}	1.69	[1.50, 1.91]	4e-17	479
counter_surprise:397_{28, 29, 34}	1.69	[1.49, 1.90]	4e-17	495
counter_surprise:397_{29, 34} (two-bin)	1.67	[1.50, 1.87]	2e-19	611
counter_surprise:397_{28, 29} (two-bin)	1.66	[1.48, 1.85]	2e-19	630

Interpretation: **sustained anomalies concentrated in the histogram- 397 bin range 27-35 double the near-term hazard of Component X repair.** The signal is a cumulative-usage-profile marker at the truck level rather than a temporal precursor in the last-K events; the temporally-held-out AUROC (§6.2) is 0.60 for this reason.

Use: **fleet triage.** Rank trucks by the count of significant histogram-397 patterns present in the last-20-readout window; prioritise workshop inspection for the top decile.

BGL (LLNL Blue Gene/L syslogs).

No non-alert precursor pattern passes the post-selection-valid BY $q < 0.05$ test on any horizon.

Interpretation: **alert cascades on BGL are self-triggering**. The non-alert stream (`system_error` , `system_warning` , `system_info`) carries no predictive information about the first alert of the next cascade. This is a different question from the one deep-learning log anomaly detectors answer on BGL (DeepLog, LogBERT, and related methods flag anomalous alert lines themselves); here the task is whether the non-alert stream predicts the first alert of the next episode, and on BGL it does not.

Use: **negative recommendation**. do not deploy this pipeline on HPC-syslog data as an alert-cascade early-warning system. A better use of pattern mining on this trace is post-hoc cascade taxonomy (which alert codes cluster together within an episode) rather than prediction.

Sections 6.7 through 6.11 present the aggregate evidence behind these signatures: the predictive-vs-frequent-noise separation (§6.7), post-selection-valid sequence significance (§6.8), the same-sample marginal baseline (§6.9), the SCANIA matched hazard-ratio analysis (§6.10), and lead time on true positives (§6.11).

6.7 Predictive vs frequent-noise separation across traces

The paper's central object is the fraction of mined frequent patterns that pass the per-pattern statistical significance test on an inference sample disjoint from the discovery sample used for mining. Frequency alone is a weak proxy for predictiveness; the post-selection-valid test tells us which mined patterns carry elevated failure signal against matched controls under an inference regime not contaminated by the label-dependent candidate selection.

Every trace is split 50/50 entity-disjoint into a discovery half (for FP-Growth candidate selection) and an inference half (for exact hypergeometric p-values and BH / BY correction).

Table 11. Post-selection-valid itemset significance by trace and horizon on the discovery/inference split.

trace	horizon	disc entities	patterns mined on disc	sig BH $q < 0.05$	sig BY $q < 0.05$
Azure	24h	50	8	8 (100%)	8 (100%)
Azure	last5	50	79	52 (66%)	46 (58%)
Azure	last10	50	815	379 (46%)	241 (30%)
Alibaba	last3	~1.12M jobs	11	6 (55%)	5 (45%)
Alibaba	last5	~1.12M jobs	11	5 (45%)	5 (45%)
Alibaba	last10	~1.12M jobs	11	5 (45%)	5 (45%)
BGL	last5	32	2	1 (50%)	1 (50%)
BGL	last10	32	3	1 (33%)	1 (33%)
BGL	last20	32	3	1 (33%)	1 (33%)
SCANIA	last5	11,775	66	9 (14%)	8 (12%)
SCANIA	last10	11,775	597	41 (7%)	6 (1%)
SCANIA	last20	11,775	37,797	0 (0%)	0 (0%)

Post-selection-valid inference has two visible consequences. First, the Azure `last10` fraction drops from 86% (naive) to 30% (BY-corrected on inference half); the extra patterns were selection artefacts. Second, the SCANIA `last20` fraction drops from 6.0% (naive) to 0% (post-selection valid): the 37,797-pattern discovery-half mining run at minimum support 0.05 does not survive an honest inference test. Both direction and magnitude of these shifts match the pre-registered concern about mining-and-testing on the same sample.

The Azure and Alibaba results persist in weaker but still substantive form (46-100% at rich horizons); SCANIA under post-selection-valid inference no longer supports an aggregate “some fraction is predictive” claim on this mining threshold and requires the matched conditional-logistic analysis in §6.10 instead. On Azure and Alibaba the majority of frequent patterns are also predictive; on BGL and SCANIA the majority are frequent-but-noise, and the paper’s concrete predictive-pattern list is short.

6.8 Post-selection-valid sequence significance and closed-sequence compression

We extend the discovery/inference split to sequences: PrefixSpan runs on the 50% discovery half, and each mined sequence is scored on the 50% inference half via an exact hypergeometric test on windows that contain the sequence as an ordered subsequence, then corrected across all sequences tested at each horizon.

Table 12. Post-selection-valid sequence significance by trace and horizon.

trace	horizon	disc windows	inf windows	sequences mined on disc	sig BH q<0.05	sig BY q<0.05
Azure	24h	1,488	1,484	7	7 (100%)	7 (100%)
Azure	last5	1,488	1,484	70	51 (73%)	51 (73%)
Azure	last10	1,488	1,484	694	537 (77%)	461 (66%)
Alibaba	last3	32,474	32,474	16	9 (56%)	9 (56%)
Alibaba	last5	32,474	32,474	29	11 (38%)	11 (38%)
Alibaba	last10	32,474	32,474	113	54 (48%)	49 (43%)
BGL	last5	14,443	14,432	10	5 (50%)	5 (50%)
BGL	last10	14,443	14,432	25	10 (40%)	5 (20%)
BGL	last20	14,443	14,432	134	4 (3%)	4 (3%)
SCANIA	last5	4,544	4,544	65	9 (14%)	8 (12%)
SCANIA	last10	4,544	4,544	548	39 (7%)	8 (1%)
SCANIA	last20	4,544	4,544	6,262	11 (0.2%)	3 (0.05%)

The sequence result mirrors the itemset finding under the same split: sequences carry a large post-selection-valid signal on Azure (77% BH-significant at `last10` , 66% BY) and a moderate one on Alibaba (48% BH at `last10`). On BGL and SCANIA the significant fraction collapses as longer horizons expand the candidate space faster than the signal grows. On SCANIA `last20` the same 6,262-sequence mining run reduces to 3 BY-significant sequences, a strong post-selection-inflation warning that mirrors the itemset case.

To measure the redundancy in the raw PrefixSpan output we also run CloSpan [52], as implemented in SPMF [9], at the same 5% minimum support: a closed sequential pattern is one whose support is strictly larger than any of its super-sequences. On Azure and SCANIA, PrefixSpan output is already tight (1:1 compression); on BGL and Alibaba, closed sequences compress the raw list by up to a factor of two (BGL `last5` 10/20, `last10` 26/39, `last20` 138/150; Alibaba `last3` 15/16). Closed sequences are the input the deployment-facing signature catalog uses.

6.9 Formal significance (same-sample marginal counts)

The same-sample BH counts (mining and testing on the full training sample) quantify the post-selection inflation that the §4.5 discovery/inference design removes: they are the naive marginal-p baseline, and the gap between them and the post-selection-valid counts in §6.7 and §6.8 is the size of the selection artefact. At BH $q < 0.05$: every Azure 24h itemset (6/6) and every Azure 24h sequence (7/7) is significant; 53/77 Azure `last5` itemsets and 55/67 Azure `last5` sequences; 562/657 Azure `last10` sequences. Both 1h and 6h Azure horizons flag zero patterns as expected (0/3 sequences at 1h, 0/5 at 6h). On Alibaba: 6/10 `last3` itemsets, 9/16 `last3` sequences, 59/109 `last10` sequences.

6.10 SCANIA risk-set matched patterns (Component X)

Applying the risk-set matched-sampling extension from §4.4 to SCANIA Component X (2,272 cases $\times$ 3 controls each drawn from the risk set at each case's failure lifetime), FP-Growth at minimum support 0.05 mines 42,453 candidate itemsets from the `counter_surprise` event stream.

We estimate the per-pattern hazard ratio via **conditional logistic regression stratified by matched risk set** (§4.6), fitted with `statsmodels.ConditionalLogit`. On the top-200 patterns by case-hit count, **117 (58.5%) pass the joint criterion HR CI excludes 1 AND BH $q < 0.05$, and 108 (54.0%) survive the stricter Benjamini-Yekutieli arbitrary-dependence FDR at $q < 0.05$** (BH $q < 0.01$: 128; BY $q < 0.01$: 117); all p-values are `ConditionalLogit` Wald p-values, corrected across the 200 patterns actually tested. Top hazard ratios cluster in feature-397 bin combinations, consistent with the concentration of predictive signal in a small number of underlying histograms.

Top 5 predictive Component X signatures (matched HR, 95% CI, p):

Table 13. Top five SCANIA Component X signatures with matched hazard ratios and 95% confidence intervals.

HR	95% CI	p	n_case	n_control	pattern
1.73	[1.53, 1.96]	1.0e-17	456	(matched)	counter_surprise:397_{27, 28, 29}
1.69	[1.50, 1.91]	4.1e-17	479	(matched)	counter_surprise:397_{29, 34, 35}
1.69	[1.49, 1.91]	3.7e-17	495	(matched)	counter_surprise:397_{28, 29, 34}
1.67	[1.50, 1.87]	2.1e-19	611	(matched)	counter_surprise:397_{29, 34}
1.66	[1.48, 1.85]	2.1e-19	630	(matched)	counter_surprise:397_{28, 29}

The **pooled 2×2 analysis previously reported inflated these effects by roughly 1.6×** (top pooled MH-OR 2.72 [2.10, 3.51] for the same pattern), because pooling discards the matched-set structure that the incidence-density sampling design creates. The matched HR of 1.73 is the correct estimator under Prentice-Breslow, and its tighter CI reflects that the matched design conditions out inter-vehicle heterogeneity the pooled analysis conflated with the pattern effect.

Top hazard ratios are concentrated in histogram-397 bin combinations. Grouping matched-significant patterns by their dominant histogram feature: **feature 397 supplies the majority of significant patterns; features 158 and 167 contribute a smaller number of independent signatures.** The “N distinct predictive patterns” reading of the matched output must therefore be interpreted with awareness of this underlying feature-set concentration.

Top 5 predictive Component X signatures (MH-OR, 95% CI):

Table 14. Pooled 2x2 odds ratios for the same SCANIA signatures, superseded by the matched estimates in Table 13.

MH-OR	95% CI	n_case	n_control	pattern (event_subtype_seq)
2.72	[2.10, 3.51]	114	130	counter_surprise:397_{10, 27, 28, 29}
2.65	[2.07, 3.39]	122	143	counter_surprise:{158_3, 397_28}
2.63	[2.05, 3.38]	118	139	counter_surprise:397_{10, 27, 29, 34}
2.44	[1.98, 3.01]	165	212	counter_surprise:397_{10, 28, 29, 34}
2.37	[1.94, 2.89]	183	243	counter_surprise:{158_9, 309_0}

The top signatures are concentrated in bin combinations of the same histogram feature (397), consistent with the histogram-encoded feature set described in §3.6. The 158_9 + 309_0 cross-feature signature is an example of a two-feature interaction that pattern mining surfaces without needing a black-box classifier. Despite these interpretable hazard-ratio-scored patterns, none of them lift a temporally-held-out logistic regression beyond AUROC 0.60 (per §6.2): the patterns are per-truck static discriminators, not temporal precursors that a next-K-event alarm can act on. The distinction is important operationally: hazard-ratio-scored patterns support cohort-level fleet triage (which trucks warrant closer inspection), not next-event alerting.

6.11 Lead time on true positives

- **Alibaba** (operational lead time): median 0 s across horizons, IQR 0-2 min. Task boundaries are the practical alarm resolution; the AUROC 0.74-0.81 signal is real but must be acted on within the next-task interval.
- **Azure** (structural lead time set by the generator): median lead time is exactly 24.0 hours at every horizon. The synthetic generator places pre-failure errors on a ~24h clock relative to the failure timestamp, so any correctly-triggered alarm inherits that clock. This describes the dataset’s construction, not the model’s early-warning capacity.

Lead-time detail: results/tables/{azure,alibaba}_leadtime.md.

7 Discussion

7.1 Is the predictor in the entire sequence or in a subpart?

Given a mined sequence S with lift $L(S)$, we ask: does any proper subsequence $S' \subset S$ in the mined set already reach lift(S)? If so, the predictor lives in the subpart and S is redundant; if no proper subsequence matches, the FULL ordered sequence is the minimal predictor. Formally, S is “full-sequence-dominant” iff $\text{lift}(S) > \text{lift}(S') + 0.05$ for every proper subsequence S' the miner also produced.

Top-200 sequences per horizon:

Table 15. Full-sequence-dominant versus subpart-dominant predictive sequences by trace.

trace	horizon	full-sequence-dominant	subpart dominant	fraction full
Azure	last5	41	11	79%
Azure	last10	191	9	96%
Alibaba	last3	2	10	17%
Alibaba	last5	5	20	20%
Alibaba	last10	31	74	30%

The finding is trace-dependent:

- **Azure PdM: predictor IS in the entire sequence.** On `last10`, 96% of top-200 predictive sequences are full-sequence-dominant, meaning no proper subsequence they contain reaches their lift. Concretely, `maintenance:comp4 → software_error:error2 → software_error:error3` reaches lift 3.73; the best proper subseq `software_error:error2 → software_error:error3` has lift 2.55 (delta +1.18). The full ordered chain adds real signal beyond any of its parts. Operationally, an alarm should be keyed on the full ordered sequence, not on any two-event fragment of it.
- **Alibaba v2018: predictor lives in a shorter subpart.** Only 17-30% of top sequences are full-sequence-dominant. Most are dominated by a short leading subsequence, often just `task_waiting:R` alone (lift ~3.98). Once a Waiting task appears in a job, the failure risk is set; adding subsequent Success events to the pattern does not lift it further. Operationally, the useful alarm is short.

This resolves an ambiguity the raw order-gain distribution left open. Order helps on both traces (§6.3), but for different reasons: on Azure the full ordering contributes signal beyond every subpart, and on Alibaba the ordering just distinguishes one privileged short prefix from bag-of-items noise.

7.2 What each mined signature means operationally

On Azure PdM, `software_error:error2 → software_error:error3` at `last5` reaches sequence lift 3.73 vs itemset lift 2.22 for the same items. The order-specific reading is that `error2` and `error3` are not interchangeable noise: a machine reporting `error2` first and then `error3` is materially more likely to reach a `terminal_failure` than one that reports them in the other order. In practical monitoring, an alarm keyed on the pair-in-order is preferable to the same alarm keyed on the pair-as-set.

On Alibaba, `task_success:M → task_success:M → task_success:M` at `last3` reaches sequence lift 3.06 vs itemset lift 1.37 for the same items. Three consecutive Map completions predict a subsequent Failed task more strongly than “the job contains Map completions” alone. The operational reading is that the position of the failure inside the DAG matters: jobs that make it through a Map-heavy prefix are the jobs whose downstream Reduce or Join phases can fail, whereas jobs that fail early do so in a different distribution of task types.

7.3 Six contrasting case studies

Six heterogeneous traces inform when pattern mining recovers meaningful predictive structure and when it does not. They differ along many axes (domain, event vocabulary, synthetic vs real, entity definition, target construction, observation cadence, class prevalence, control sampling), so we describe the pattern rather than claim a causal rule. The strongest positive evidence is the two-farm wind cascade: two independent Senvion fleets at different sites produce the same lift-4.00 zero-control cascade signatures under one protocol, which is the closest the survey comes to a controlled replication.

The case studies answer a natural follow-up question: the combined-feature-set advantage does not transfer to every operational event log.

- **Where the method works cleanest** (Kelmars, Penmanshiel wind farms): a genuine physical cascade. Continuous mechanical wear produces discrete vendor alarm codes at intermediate stages, and a specific alarm terminates the cascade as a Forced outage. The precursor codes (generator-fan overload 2550/2650/2655 on Kelmars; safety-system 9000/9210 and frequency-converter 3000 on Penmanshiel) fire in inference-half failure windows and never in matched controls at `last5 / last10`, and the mined-itemset rule strictly dominates a most-recent-event baseline (§6.5). The two farms cross-validate each other: different rotor sizes (MM92 vs MM82), different sites, same protocol, same lift-4.00 cascade shape.
- **Where the method works** (Azure PdM, Alibaba v2018): rich native discrete event vocabularies (5 error codes $\times$ 4 component types on Azure; 4 task statuses $\times$ 6 task roles on Alibaba). Failure-window content differs discriminably from control-window content, and order carries information beyond the itemset.
- **Where the method does not work** (BGL, SCANIA Component X): the target class is self-triggering (BGL alerts follow other alerts, and non-alert log lines carry no discriminable precursor signal), or the discrete event stream must be derived from continuous counters (SCANIA requires binning per-vehicle deltas, and a defensible 90th-percentile surprise binning produces only marginal AUROC).

Concretely: on BGL the best combined AUROC across horizons is 0.51 (chance) even when INFO-level messages and component granularity are included in the non-alert stream; on SCANIA the best combined AUROC across horizons is 0.60, unchanged when the fleet-wide 90th-percentile delta binning is replaced by per-vehicle-normalised 90th-percentile binning.

To determine whether the SCANIA gap is representation-loss (tokens destroy signal that is actually in the trace) or signal-absence (no representation could recover 0.75), we run a diagnostic ceiling test: LightGBM on a compact set of histogram-aware distributional descriptors (Wasserstein-1 distance to a per-vehicle baseline, signed centroid shift, entropy shift, tail-mass shift, and each descriptor's slope over the last 20 readouts) computed on the same temporal split. The ceiling model reaches AUROC 0.60 / AUPRC 0.04, essentially identical to the pattern-mining pipeline. Logistic regression on the same 113 structured features reaches 0.58 / 0.05. The GBM-versus-LR gap is 0.02 AUROC, so classifier capacity is not the constraint either. The gap between SCANIA (~0.60) and the Azure / Alibaba wins (~0.80-1.00) reflects a limit of the readout cadence and feature vocabulary, not of the pattern-mining pipeline against a richer alternative representation.

Positive control on same-manufacturer data. To rule out the alternative explanation that SCANIA-family telemetry itself lacks predictive signal, we apply the same LightGBM ceiling test to SCANIA APS Failure at Scania Trucks [39] (UCI 421, IDA 2016 industrial challenge). APS Failure uses the same anonymised

histogram-encoded schema as Component X (7 histogram groups of 10 bins each + 100 single counters = 170 feature columns) but delivers one per-truck cross-sectional readout instead of a longitudinal readout stream, with a binary APS-system-failure label at a 1.67% positive rate. On the same LightGBM configuration our Component X ceiling used, APS Failure reaches AUROC 0.994 / AUPRC 0.934 on the canonical held-out test split (16,000 trucks, 375 positives); LR alone reaches 0.979 / 0.800. Same manufacturer, same anonymisation schema, different readout format, near-perfect predictability. This excludes “SCANIA-family data is inherently weak” as an explanation for the Component X boundary.

Root-cause diagnosis: trajectory-signal absence, not signal absence. A stratified 5-fold cross-validation on the same 420 aggregated Component X features (mean / max / std / last of each of the 105 columns per vehicle), with each column linearly residualised against `length_of_study_time_step` inside every fold, reaches AUROC 0.826 ± 0.005 , comparable to Alibaba’s `last3` combined score (0.81) and BGL’s chance (0.51). The same LightGBM configuration under the temporal split used elsewhere in the paper only reaches 0.67, and the pattern-mining pipeline reaches 0.60. Component X features therefore carry substantial per-truck failure signal, but the signal is a static per-vehicle profile (aggregate usage, cumulative counter shape) rather than a temporal degradation trajectory. Last-K-events windows and the ordered-pattern mining built on them cannot see it, because there is no pre-failure event ordering to catch; the discriminative information is spread across the truck’s entire operating history.

This gives a sharper three-way regime-of-validity: (i) four positive traces (Kelmars, Penmanshiel, Azure PdM, Alibaba v2018), where target failure is preceded by a discriminable ordered event trajectory, with the two wind farms the cleanest physical-cascade instances; (ii) BGL, where the target class is self-triggering with no discriminable non-alert precursor; (iii) Component X, where the target has strong per-truck signal in aggregate features but no last-K-events trajectory signal, so pattern mining on windows attains the temporal-split ceiling but not the transductive per-vehicle ceiling. The APS positive control confirms that (iii) is a target-shape distinction, not a manufacturer or schema deficiency.

The method’s regime of validity is therefore “trace has a rich native discrete event vocabulary AND failure class is not self-triggering AND readout-cadence signal capacity exceeds the target AUROC bar”. BGL fails the second condition; SCANIA fails the third; the two wind farms, Azure PdM, and Alibaba v2018 satisfy all three. The wind farms are the strongest members of the positive group because the cascade is literally physical: a mechanical wear process generates the intermediate alarm codes that the mining recovers.

The two lead-time regimes in §6.11 speak to deployment. Azure inherits a structural 24h clock from the synthetic generator and should not be read as a real-world warning interval; Alibaba’s median 0-second lead time is the honest one, and a per-job classifier there must be paired with sub-second scheduling infrastructure to act on the signal at all.

8 Limitations

- Alibaba results are per-job, computed on `batch_task` alone; the `batch_instance` table (21 GB compressed) that would enable per-machine failure trajectories on the same trace is left to future work.
- Alibaba sequence patterns are numerically fewer than Azure ones (2-6 vs 6-16 surviving the shuffle-null per horizon). A wider minimum support sweep and top-K sequence mining would sharpen the Alibaba sequence-mining slice specifically.

- The minimum support sensitivity sweep covers itemset support; regularization strength and cross-machine / cross-job leave-one-out are the next robustness questions.
- Azure lead times are set by the synthetic data generator and describe dataset construction rather than real-world warning intervals. Alibaba lead times are the operationally-honest number from a real trace.
- SCANIA binning uses per-vehicle 90th-percentile deltas; a stronger alternative representation (Wasserstein-1 distance to vehicle baseline, centroid shift, entropy and tail-mass changes per histogram feature) was also tested via a LightGBM ceiling diagnostic and reached the same AUROC 0.60. The signal at this readout cadence is insufficient to reach the 0.75 bar under any representation we tested; scoping SCANIA as a boundary case reflects that finding.

Scope of the statistical guarantees. The discovery/inference split is a single seeded partition; the reported fractions are point estimates under that partition, and the fixed seed policy is stated in §9. FDR is controlled within each (horizon, pattern-class) family, so a reader comparing significant fractions across horizons is reading several families rather than one corrected sweep. The inference-half lift is bounded above by 4.0 under 1:3 case-to-control matching, so a lift of 4.00 denotes a pattern that is present in failure windows and absent from every matched control, not an unbounded effect size. The cross-farm replication is cross-site and cross-model (MM92 vs MM82) but within one vendor's alarm firmware (Senvion), so it establishes transfer across turbine model and site rather than across alarm vocabularies. The Penmanshiel window covers the site's 2016 commissioning phase, which the safety-system 9000/9210 family reflects. The operational base rate of 0.01 used for the calibrated PPV in §6.6 is an assumed deployment prior, not a measured fleet rate, and the PPV scales with it.

9 Reproducibility

All code, mined-pattern parquets, discovery/inference splits, matched hazard-ratio outputs, and end-to-end reproduction scripts are released at the paper's GitHub repository (<https://github.com/ApartsinProjects/FailurePatterns>) with the rendered manuscript hosted at <https://apartsinprojects.github.io/FailurePatterns/>. The repository carries: (i) per-dataset ingest scripts referencing the exact public sources for Azure PdM (Kaggle mirror), Alibaba cluster-trace-v2018 (Alibaba OSS `batch_task.tar.gz`, ~130 MB), BGL (Loghub Zenodo `8196385/BGL.zip`, ~55 MB), SCANIA Component X (Swedish National Data Service DOI `10.5878/jvb5-d390`, CC-BY-4.0), Kelmarsh (Zenodo `10.5281/zenodo.5841834`, CC-BY-4.0) and Penmanshiel (Zenodo `10.5281/zenodo.5946808`, CC-BY-4.0) SCADA status logs; (ii) window construction, mining (mlxtend FP-Growth 0.25.0 for itemsets, SPMF 2.64 via subprocess for PrefixSpan sequences), scoring, discovery/ inference splitting, and matched conditional-logistic scripts; (iii) all reported numbers programmatically verified by `scripts/audit_paper_numbers.py` against the artefact JSON/parquet files (numbers audit: 100% of claims pass at last release); (iv) `scripts/publish_paper.sh` regenerates the manuscript HTML and DOCX from Markdown source with pandoc + citeproc + the bibliography, and copies the HTML into `docs/index.html` for GitHub Pages.

Python 3.14 pinned via `requirements.txt`; Java 21 required for SPMF invocation. Random seeds fixed at 20260828 for every stochastic step (discovery/inference split, control sampling, permutation tests, LightGBM). Wall-clock, directly measured by `scripts/measure_reproducibility_walltime.py` and saved to `results/patterns/reproducibility_timing.json`: **the downstream pipeline (itemset mining, sequence mining, post-selection split, matched conditional logistic, closed-sequence CloSpan, wind-**

farm signature extraction) totals 36.9 minutes on a single Windows CPU- only workstation across all six traces. SCANIA itemset mining is the dominant single stage at 26.2 minutes (counter_surprise event stream, 4,544 windows x 42k candidate itemsets); the two wind-farm signature runs together take 8.6 seconds. Raw-trace download and initial window construction are one-time steps documented in the ingest_*.json artefacts and excluded from this number.

Data availability and ethics. All six datasets are public and openly licensed (the four cited above under CC-BY-4.0, plus the Azure PdM sample and Alibaba cluster-trace-v2018 under their respective public terms); the paper introduces no new data collection. None of the traces contains personal or human-subject data: they are machine telemetry, cluster job logs, HPC syslogs, and industrial alarm codes. The derived event streams, windows, mined patterns, and all result artefacts are released in the repository under the same open terms.

10 Conclusion

Frequent-pattern mining of discrete operational events surfaces interpretable pre-failure signatures on four of six traces studied. The cleanest evidence is a physical cascade replicated across two independent Senvion wind farms: on Kelmarsh and Penmanshiel, mining recovers zero-control cascade signatures at inference-half lift 4.00, where a mechanical wear process generates the intermediate alarm codes that terminate as Forced outages, and the mined-itemset rule strictly dominates a most-recent-event baseline. On the two cloud / maintenance traces (Azure PdM, Alibaba v2018), sequences add real predictive information beyond itemsets when window definitions are rich enough for order to be a real degree of freedom; at those horizons, combining itemset and sequence features improves failure prediction by 5-10 AUROC points over either alone. The method's regime of validity is mapped by two boundary traces (BGL, SCANIA) where the pipeline does not find last-K-events signal, with a mechanistic explanation for each, and by a matched conditional- logistic analysis that recovers interpretable per-truck hazard ratios on SCANIA under right censoring.

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